

Derivation of the GDI linear matrix L in `jax_rmhd` and resolution of the apparent discrepancy with eq. (5.3) of the notes

P4a implementation note

2026-08-01

All equation numbers refer to `docs/GDI_nonlinear_equations (10).pdf`. The code evolves $f = (N, \varphi)$ with the convention $\partial_t f = Lf + \mathcal{N}(f)$, propagator $e^{L\tau}$ (putzer2 backend). Everything below is in the normalized units of §5.1: $\rho_s = c_s = \Omega_i = 1$.

1 Starting point

The normalized 2D equations (5.4)–(5.5), with the parallel current-closure term represented by a perpendicular- k -local $\gamma_{\parallel}$:

$$\frac{\partial N}{\partial t} - \frac{1}{L_n} \frac{\partial \varphi}{\partial y} + \{\varphi, N\} - \gamma_{\parallel} [\varphi - N] = 0, \quad (5.4)$$

$$\frac{\partial \nabla_{\perp}^2 \varphi}{\partial t} + \{\varphi, \nabla_{\perp}^2 \varphi\} + \nu_{in} \nabla_{\perp}^2 \varphi - \nu_{in} v_0 \frac{\partial N}{\partial y} - \gamma_{\parallel} [\varphi - N] = 0. \quad (5.5)$$

For P4a (strictly 2D) the closure floor of eq. (4.3) is used, scaled by the tunable `eqpars` key `gpar_fac`:

$$\gamma_{\parallel} = \text{gpar_fac} \, \nu_{in} k_{\perp}^2 \text{ (default 1)}. \quad (1)$$

(Relation to eq. (4.7)’s α : with $\gamma_{\parallel}$ substituted for $D_{\parallel} k_z^2$ one gets $\alpha = 1 + \text{gpar_fac}$, so the default reproduces $\alpha = 2$, the minimum- $\gamma_{\parallel}$ factor quoted at eq. (4.6). Below, α denotes `gpar_fac` itself only inside the matrix (4), where it abbreviates $\gamma_{\parallel}/k_{\perp}^2 \nu_{in}$.) Note $\gamma_{\parallel}/k_{\perp}^2 = \alpha \nu_{in}$ is k -independent under this closure — this constant appears repeatedly below.

2 Fourier form and the vorticity division

With $\partial_y \rightarrow ik_y$ and $\nabla_{\perp}^2 \rightarrow -k_{\perp}^2$, the linear part of (5.4) is immediate:

$$\partial_t N = \underbrace{-\gamma_{\parallel}}_{L_{NN}} N + \underbrace{\left(\gamma_{\parallel} + \frac{ik_y}{L_n}\right)}_{L_{N\varphi}} \varphi + (\text{bracket}). \quad (2)$$

Equation (5.5) evolves the vorticity $-k_{\perp}^2 \varphi$; dividing by $-k_{\perp}^2$ (zero mode masked via `inv_ksq`, exactly as the RMHD φ equation):

$$\partial_t \varphi = \underbrace{\left(\frac{\gamma_{\parallel}}{k_{\perp}^2} - \frac{ik_y \nu_{in} v_0}{k_{\perp}^2}\right)}_{L_{\varphi N}} N + \underbrace{\left(-\nu_{in} - \frac{\gamma_{\parallel}}{k_{\perp}^2}\right)}_{L_{\varphi \varphi}} \varphi + \frac{\{\varphi, \nabla_{\perp}^2 \varphi\}}{-k_{\perp}^2} \Big|_{\text{sign as coded}}. \quad (3)$$

Collecting, and adding the numerical hyperdissipation closure on both diagonal entries (required because the linear growth rate does not decay at large $k_\perp$, cf. eq. 3.14):

$$L = \begin{pmatrix} -\gamma_\parallel - D_k & \gamma_\parallel + \frac{ik_y}{L_n} \\ \alpha\nu_{in} - \frac{ik_y\nu_{in}v_0}{k_\perp^2} & -\nu_{in} - \alpha\nu_{in} - D_k \end{pmatrix}, \quad D_k \equiv \text{diss } k_\perp^{2\text{hyper}}, \quad (4)$$

using $\gamma_\parallel/k_\perp^2 = \alpha\nu_{in}$ from (1). This is verbatim `_L_entries` in `jax_rmhd/physics/gdi.py`. (The ik_y factors use k_y with the Nyquist row zeroed: an odd-order spectral derivative is ill-defined on the self-conjugate row, and unlike the nonlinear terms, L is applied without dealiasing, so the row must be fixed explicitly to preserve rfft2 reality. One edge row, no interior physics.)

3 Check against the dispersion relation

With $f \propto e^{\lambda t}$ and $D_k = 0$, the characteristic polynomial of (4), $\lambda^2 - (\text{tr } L)\lambda + \det L = 0$, factors as

$$(\lambda + \gamma_\parallel) \left[(\lambda + \nu_{in})k_\perp^2 + \gamma_\parallel \right] = (\gamma_\parallel + i\omega_*)(\gamma_\parallel - i\omega_{\text{Ped}}), \quad \omega_* \equiv \frac{k_y}{L_n}, \quad \omega_{\text{Ped}} \equiv k_y\nu_{in}v_0. \quad (5)$$

Substituting $\lambda = -i\omega$ (each bracket picks up a factor $-i$; the right side rearranges with $i \cdot (-i) = 1$) gives exactly eq. (3.7) of the notes:

$$(\omega + i\gamma_\parallel) \left[(\omega + i\nu_{in})k_\perp^2 + i\gamma_\parallel \right] = -(\omega_{\text{Ped}} + i\gamma_\parallel)(\omega_* - i\gamma_\parallel). \quad (3.7)$$

At $\gamma_\parallel = 0$ this collapses to $\omega(\omega + i\nu_{in}) = -\omega_*\omega_{\text{Ped}}/k_\perp^2 = -A$ with $A = k_y^2\nu_{in}v_0/(k_\perp^2 L_n)$, i.e. eq. (2.8); the code's tests further verify the collisional limit (2.9), the inertial limit (2.11), and, at finite $\gamma_\parallel$, the roots of (3.7) directly (numpy `roots` vs. eigenvalues of (4), `rtol` 10^{-10}), plus a propagator-level check that putzer2-evolved fields grow at $\max \text{Re}(m \pm s)$.

4 The “corrected term” that wasn’t

The P4a implementation report claimed the derived $L_{\varphi N}$ contains an $\alpha\nu_{in}$ term *absent* from the notes’ transcribed matrix (5.3). That claim is wrong, and the note-writer (not the paper) needed correcting: on visual inspection of p. 7, eq. (5.3) reads, in the $\partial_t f + Mf = 0$ convention,

$$M = \begin{pmatrix} \gamma_\parallel & -ik_y\rho_s c_s/L_n - \gamma_\parallel \\ ik_y\nu_{in}v_0/k_\perp^2\rho_s c_s - \gamma_\parallel/k_\perp^2\rho_s^2 & \nu_{in} + \gamma_\parallel/k_\perp^2\rho_s^2 \end{pmatrix}, \quad (5.3)$$

whose (φ, N) entry *does* include $-\gamma_\parallel/k_\perp^2\rho_s^2$. Hence $L = -M$ agrees with (4) in *every* entry (before the added hyperdissipation): the derivation from (5.4)–(5.5), the transcription (5.3), and the implemented matrix are all mutually consistent. The implementation’s validation never relied on (5.3) — the dispersion-relation tests above are the arbiter — but no discrepancy with the notes exists.

5 Sanity limits

$\gamma_{\parallel} \rightarrow \infty$ (adiabatic): the N -equation row forces $N \rightarrow \varphi$ (eq. 3.8 with $\gamma_{\parallel}$ dominant), and the growth shuts off — consistent with the stabilization role of current closure. ν_{in} enters $L_{\varphi\varphi}$ as bare vorticity friction with relaxation rate exactly ν_{in} (the “mild” stiffness claim in the plan’s design background). Growing solutions at $A > 0$: $s^2 = m^2 - \det L$ is complex in general; all propagator arithmetic is complex, and the growth rate extracted from evolved fields matches $\text{Re}(m + s)$ to 10^{-6} (fp64).